



TesseraApp

PRESENTATION & DEMO GUIDE

What to Present, and in What Order

The run sheet for the Final Presentation and its live demo, plus the Implementation 2 code walkthrough — slide-by-slide talking points, demo scenarios, timings, and safety nets.

Final Presentation · ~18 min

Live demo · 2 scenarios

Implementation 2 · code walkthrough

PRESENTERS

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INSTITUTION

Lewis University

COURSE · ADVISOR

CPSC 69100 · Prof. Larry Motuzis

DELIVERY

Record → upload to YouTube → submit link

Universal setup checklist

Applies to the presentation, its live demo, and the code walkthrough. This is a **guide** — follow the order and adapt the talking points to your own voice.

1080p

RECORD RESOLUTION

Headset

MIC + QUIET ROOM

Rehearse

ONCE, END TO END

- **Prep the app:** run `./start.sh`, confirm the DB is seeded, and log out so you begin clean. Have an authenticator app ready.
- **Legibility:** increase browser / editor font size so UI and code read clearly on playback.
- **Tabs ready:** the running app (`localhost:4200`), the architecture diagram, and your IDE.
- **Show, don't read:** spend time clicking real flows, not narrating screenshots.
- **Be honest about limitations** — it reads as maturity, not weakness.
- **Backup:** keep a screen-recording of each demo scenario in case the live demo misbehaves.

DEMO ACCOUNTS

Log in as `eve.admin@tessera.dev` / `TesseraDemo@1` (admin). Other seeded roles share the same password — see the Final Report, Appendix A.

01 Live demo — full run order

This is the working-software portion — the part that must succeed on camera. Run it inside the Final Presentation (slides 17–21) or standalone. Prove the project works, feature by feature.

~TIME	SHOW
1.5 min	A • Registration & verification — register a new account; note it starts <i>disabled</i> and activates via an email link (in dev, shown in the server console).
2 min	B • Login + MFA — log in, complete the authenticator challenge, enter a live code, land on the dashboard. Mention the brute-force throttle; TOTP supersedes SMS.
2.5 min	C • Security Center — enroll an authenticator (QR → confirm → recovery codes); show the sessions/devices panel; revoke a session; one line on reuse detection.
2 min	D • Administration & RBAC — search Users; open a user, reassign their role; show it in their audit history; mention org scoping.
1.5 min	E • Business domain — create a customer; create/attach an invoice; export an XLSX report.
1 min	F • Federated login — (if provider credentials are configured) Continue with Google → callback → our JWT; the federated session shows in the device list.

TIMING SAFETY NET

Running long? Trim **E** and the registration detail in **A**. The security features (**B**, **C**, **D**) are the differentiators — do not cut them. If federation isn't live yet, present **F** as a short walkthrough/roadmap rather than a live click-through.

02 Slide-by-slide run sheet

Narrated deck (26 slides) + a live demo of the scenarios above. **Organization and style are graded** — one idea per slide, talk to the audience, and let the demo prove it works. Slide numbers match the branded *Final Presentation* deck.

SLIDES	TALKING POINT	TIME
1 Title	Greeting, both names, project, one-line pitch	0:30
2–3 Problem · Objectives	Why identity is hard + costly; the 5 objectives	1:30
4–5 Build-vs-buy · Hybrid	The spectrum; where TesseraApp sits; why hybrid tokens	1:30
6 Requirements	SRS highlights — don't read every line	0:45
7–8 Architecture · Pipeline	Walk the three-tier diagram; name the 4+1 views; the seven gates	1:45
9–11 Security model	The differentiator: stateless+stateful, JWT, rotation + reuse detection	2:30
12–14 MFA · RBAC · Audit	Three ways in → one JWT; permission strings; event-driven audit	1:45
15–16 Stack · Highlights	30-sec stack skim; 3–4 engineering wins	1:00
17–21 Demo	Switch to the live app — run scenarios 1–2 (add 3–4 if time)	4:30
22–24 Results · Challenges · Future	What works; lessons; honest gaps + roadmap	2:00
25–26 Conclusion · Thanks	Restate the narrow claim; invite questions	1:00

Total ≈ 18 min. Tighter limit? Trim slides 4–5 and 13–14, and run only demo scenarios 1–2. The deck's speaker-notes carry these talking points per slide.

Demo scenarios — the part that must work on camera

SCENARIO 1 · FULL AUTH JOURNEY + SECURITY CENTER · ~2.5 MIN

1. Register a new account; note it starts disabled; follow the verification link (dev: server console) to activate.
2. Log in as eve.admin@tessera.dev / TesseraDemo@1; complete the authenticator challenge.
3. Open **Security Center**: enroll an authenticator (QR → confirm → recovery codes); show sessions/devices; revoke one; explain reuse detection in a sentence.

SCENARIO 2 · ADMIN RBAC + AUDIT · ~2 MIN

1. Open **Users**; search the directory.
2. Reassign a user's role; show it landing in that user's **audit history**.
3. Mention **organization scoping** — org admins are limited to their org; full admins are not — and that a user can't change their own role.

SCENARIO 3 · CUSTOMERS & INVOICES · ~1.5 MIN

1. Create a **customer** record; open it.
2. Create an **invoice** with line items and attach it to the customer.
3. **Export** customers/invoices to Excel — the business domain the identity layer protects.

SCENARIO 4 · FEDERATED LOGIN · ~1 MIN (IF CONFIGURED)

1. Click **Continue with Google / GitHub / Microsoft**.
2. Provider redirect → authorization-code exchange → find-or-create → our own JWT.
3. Show the federated session in the **device list**, identical to an in-house one. If credentials aren't live yet, present this as a short walkthrough.

Delivery tips (organization & style)

- **One idea per slide**; minimal text; talk *to* the audience, not the screen.
- **Show, don't tell**: spend demo time clicking real flows, not narrating screenshots.
- Rehearse the transition into and out of the live demo so it's seamless.
- **Keep a screen-recording** of each scenario as a backup in case the live demo misbehaves.

03 Implementation 2 – Code Walkthrough

The required video that **explains the code**, not just the running app. Show the IDE, walk the important files, and connect each to what the user sees.

BOOKMARK THESE FILES FIRST

SecurityConfig · CustomAuthFilter · TokenProvider · SessionServiceImpl · UserController · schema.sql · app.config.ts · token.interceptor.ts · app.routes.ts

TIME	SEGMENT	WHAT TO SHOW
0:00-1:30	Intro & big picture	The 3 tiers + the hybrid auth idea (architecture diagram)
1:30-3:30	Structure tour	Backend package layout + frontend folder layout
3:30-9:30	Backend walkthrough	The request lifecycle + the security seams
9:30-13:00	Data + frontend	schema.sql / JdbcTemplate; interceptor + guards
13:00-15:30	Build/deploy + wrap	Dockerfile, CI/CD, limitations, conclusion

Backend walkthrough – trace one request & the security seams

1. SecurityConfig — ordered authority rules, STATELESS sessions, CSRF off; CustomAuthFilter registered before the username/password filter.
2. CustomAuthFilter — shouldNotFilter (public routes) and doFilterInternal; authenticates **only** when the token has authorities.
3. TokenProvider — createAccessToken vs createRefreshToken (claims authorities, sid, jti); isValidToken (expiry + passwordChangedAt).
4. SessionServiceImpl — issueTokenPair and rotate; explain **reuse detection** and why rotate is intentionally not @Transactional.
5. UserController.login — the MFA branch (TOTP vs SMS vs direct) and the HttpResponse envelope.

TIE CODE TO BEHAVIOUR

After explaining the filter, switch to the app and show a **401** → **silent refresh** in the browser dev-tools network tab. Connect code → behaviour at least twice (login, and a 401 refresh) so it's clearly a working system.

AI Tool Disclosure

AI tools, specifically Claude (Anthropic), were used to assist with drafting, structuring, and formatting this and other course deliverables throughout this project, as well as with code generation for select implementation areas, including security-related features. All architectural decisions, system design choices, implementation direction, and academic analysis reflect our own thinking and direction. We are disclosing this use proactively, in accordance with Lewis University's Academic Integrity policy.

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COURSE

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